Problem M. XOR Sum

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

Louis loves integers. He defines the value of a sequence of non-negative integers $A = [a_1, a_2, \ldots, a_k]$ as the following formula ($\oplus$ means bitwise exclusive-or):

$$\sum_{i=1}^{k} \sum_{j=1}^{i-1} a_i \oplus a_j$$

He wants to know how many different sequences A holds the following conditions:

- The length of A is k.
- The value of A is n.
- $0 \le a_i \le m$ $(1 \le i \le k)$.

Two sequences $[a_1, \ldots, a_k]$, $[b_1, \ldots, b_k]$ are considered different if and only if there exists an index i such that $a_i \neq b_i$. Tell Louis the answer module $10^9 + 7$.

Input

The input contains of single line containing three integers n, m, k, ($0 \le n \le 10^{15}$, $0 \le m \le 10^{12}$, $1 \le k \le 18$).

Output

Output the answer modulo $10^9 + 7$.

Examples

standard input	standard output
6 2 3	12
30 6 5	1520

Note

In the first example, vaild sequences are

$[0, 1, 2], [0, 2, 1], [1, 0, 2], [2, 0, 1], [1, 2, 0], [2, 1, 0], [1, 1, 2], [1, 2, 1], [2, 1, 1], [2, 2, 1], [2, 1, 2], [1, 2, 2]$.